

KRISHNA Koushik Thokala

Austin, TX | (317)-840-3830 | krishnakoushikthokala@gmail.com | [linkedin.com/in/krishna3103](https://www.linkedin.com/in/krishna3103) | github.com/krishna310301

Summary

Cloud Infrastructure Engineer with 2+ years of production NOC experience owning high-severity incidents, SLA troubleshooting, RCA, and carrier-grade reliability. Hands-on AWS/EKS, Terraform, Kubernetes, CI/CD, and observability projects focused on service health, secure automation, and cloud incident recovery.

Technical Skills

Cloud & AWS: VPC, IAM, EC2, S3, Lambda, API Gateway, EventBridge, CloudWatch, SNS, DynamoDB, CloudFront, EKS, ECR, RDS, ALB, Secrets Manager, Bedrock
Infrastructure & DevOps: Terraform, Docker, Kubernetes, Helm, Git, GitHub Actions, CI/CD, Linux, Ingress, ConfigMaps, Secrets, Probes, HPA, k6
Observability & SRE: CloudWatch Logs/Metrics/Alarms, Incident Response, RCA, Runbooks, SLA Troubleshooting, MTTR, SLO Metrics, Production Operations
Networking: TCP/IP, BGP, OSPF, MPLS, VPN, DNS, Load Balancing, Firewalls, Juniper/Cisco Routing
Programming & Databases: Python, Bash, SQL, PostgreSQL, REST APIs, FastAPI, React | **Certification:** AWS Certified Cloud Practitioner (2026)

Experience

Tata Communications

June 2022 – July 2024

Senior Engineer, Shift Lead (production NOC operations)

Pune, India

- Led 5-person global NOC shifts through 40+ daily priority incidents, coordinating triage, escalation, vendor/TAC support, field engineer dispatch, RCA notes, and cross-functional handoffs.
- Restored high-priority carrier services within 4-hour SLA targets by driving outage bridges, segment-wise fault isolation, customer-premises testing, and follow-up through service recovery.
- Diagnosed BGP/MPLS routing failures, TCP/IP connectivity issues, DWDM optical faults, and equipment failures across Juniper, Huawei, Ciena, and Alcatel platforms using alarms, logs, interface status, and loop testing.
- Owned AVP/VP-level escalations for Tier-1 telecom clients in 24/7 operations; earned Certificate of Excellence for reliable incident leadership and service restoration.

Projects

CloudOps SRE Platform | EKS, Terraform, Helm, RDS, GitHub Actions

June 2026

- Built an EKS-based SRE operations platform for service health, deployment history, incident timelines, MTTR, SLO tracking, and error-budget views across cloud operations workflows.
- Deployed frontend/backend workloads behind ALB Ingress with Helm-managed Kubernetes resources, liveness/readiness probes, resource limits, Alembic migrations, and Secrets Manager-backed database credentials.
- Added production hardening with structured JSON logs, request correlation IDs, Kubernetes manifest validation, Terraform security/cost checks, CloudWatch logs, HPA/k6 validation, and RDS secret-rotation runbooks.

CloudOps Uptime Monitor | Lambda, DynamoDB, API Gateway, Terraform

May 2026

- Architected a serverless uptime monitor with EventBridge scheduled checks, Lambda handlers, DynamoDB history, SNS alerts, CloudWatch metrics, and a CloudFront-hosted React dashboard.
- Reduced current-status lookup scope from 86,400 retained history records to 10 latest-status rows for a 10-URL, 30-day workload by redesigning the DynamoDB access pattern.
- Added state-change alerts, recovery notifications, API key usage plans, throttling, restricted CORS, KMS encryption, DLQs, 9 custom CloudWatch metrics, and CI quality gates.

AWS Incident Triage Pipeline | Lambda, CloudWatch, SNS, Bedrock, Terraform

May 2026

- Built an event-driven incident triage workflow where CloudWatch alarm state changes trigger Lambda to parse alert context, calculate deterministic severity, and publish SNS notifications.
- Integrated Amazon Bedrock for remediation-focused summaries with fallback handling; provisioned EC2, Lambda, SNS, IAM, and CloudWatch alarms via Terraform with unit tests, Bandit, and Checkov CI checks.

Education

Indiana University Bloomington

August 2024 – May 2026

Master of Science in Computer Science, GPA: 3.5/4.0

Bloomington, IN

SRM Institute of Science and Technology

June 2018 – June 2022

Bachelor of Technology in Computer Science and Engineering, CGPA: 8.4/10

Chennai, India